

AthenaK CGL–Landau-Fluid STS Validation Note

April 24, 2026

1 Purpose

This note documents the AthenaK CGL–Landau-fluid (CGL–LF) heat-flux update implemented on branch `CGL–STS–LF`. The goal is not to reproduce the full nonlinear turbulence campaign of Squire et al. (2023), but to show that the implemented method follows the same finite-volume variable strategy and can reproduce the method-level checks, including a one-dimensional oblique linear-wave setup using the Figure 17 background state.

2 Conserved Variables

The hyperbolic CGL system evolves total energy and the conserved anisotropy variable

$$A = \rho \log \left[\frac{p_{\perp}}{p_{\parallel}} \frac{\rho^2}{|B|^3} \right]. \quad (1)$$

The legacy array slot is still named `IMU`, with `IAN` used as an alias in CGL code. The Riemann solve remains in `A`, which avoids the small- $|B|$ pathologies associated with evolving magnetic moment directly.

The LF STS operator temporarily changes only the interpretation of the `IAN` slot:

$$A \rightarrow \mu_B = \frac{p_{\perp}}{|B|}. \quad (2)$$

The standard conserved-variable STS update then advances (E, μ_B) with the LF flux divergence. Immediately after the RKL2 stage, the EOS converts $\mu_B \rightarrow A$, and the normal CGL conserved-to-primitive recovery restores $(p_{\parallel}, p_{\perp})$.

3 Landau-Fluid Closure

The implemented closure uses the Snyder–Hammett–Dorland 3+1 heat fluxes,

$$q_{\parallel} = -\chi_{\parallel} \rho \nabla_{\parallel} \left(\frac{p_{\parallel}}{\rho} \right), \quad (3)$$

$$q_{\perp} = -\chi_{\perp} \left[\rho \nabla_{\parallel} \left(\frac{p_{\perp}}{\rho} \right) - p_{\perp} \left(1 - \frac{p_{\perp}}{p_{\parallel}} \right) \frac{\nabla_{\parallel} |B|}{|B|} \right], \quad (4)$$

with the finite-collision Squire et al. coefficients

$$\chi_{\perp} = \frac{2c_{\parallel}^2}{\sqrt{2\pi} c_{\parallel} |k_{\parallel}| + \nu_{\text{eff}}}, \quad \chi_{\parallel} = \frac{8c_{\parallel}^2}{\sqrt{8\pi} c_{\parallel} |k_{\parallel}| + (3\pi - 8)\nu_{\text{eff}}}. \quad (5)$$

Here ν_{eff} is the background collision frequency plus any active mirror/firehose limiter scattering at the face. In the collisionless limit, these reduce to $\chi_{\perp} = \sqrt{2/\pi} c_{\parallel}/|k_{\parallel}|$ and $\chi_{\parallel} = \sqrt{8/\pi} c_{\parallel}/|k_{\parallel}|$. Here `lf_k_parallel` is the closure wavenumber $|k_{\parallel}|$, not a conductivity. In local mode, $c_{\parallel} = \sqrt{p_{\parallel}/\rho}$ is computed from the live face state. In background mode, only this prefactor is frozen to `lf_c_parallel0`; all gradients and thermodynamic variables remain live.

The face fluxes coupled into STS are

$$F_E^i = \hat{b}^i \left(q_{\perp} + \frac{1}{2} q_{\parallel} \right), \quad F_{\mu_B}^i = \hat{b}^i \frac{q_{\perp}}{|B|}. \quad (6)$$

4 Heat-Flux Cap

Following Squire et al. (2023), equations 3.1–3.2, the unlimited heat flux q_L is capped as

$$q = \frac{q_L q_{\max}}{q_{\max} + |q_L|}. \quad (7)$$

The implemented caps are

$$q_{\perp, \max} = \sqrt{\frac{2}{\pi}} c_{\parallel} p_{\perp}, \quad q_{\parallel, \max} = \sqrt{\frac{8}{\pi}} c_{\parallel} p_{\parallel}. \quad (8)$$

The cap is applied after computing the unlimited flux with the collision-aware coefficient, so collisions and limiter scattering suppress the flux before the smooth free-streaming bound is applied.

5 Validation Workflow

The quantitative validation is generated by running `scripts/run_cgl_lf_validation.sh` from the repository root. The script runs the CGL–LF unit-style inputs, enables machine-readable diagnostic output from the problem generator, writes logs under `docs/figures/logs`, writes CSV diagnostics under `docs/figures/data`, regenerates the figures in this note, and rebuilds this PDF when `pdflatex` is available.

Each test prints the measured value, reference value or error norm, tolerance, and pass/fail status. The plotted points are read from the same diagnostic CSV files produced by the passing runs.

6 Validation Problems and Tolerances

7 Quantitative Results

Input	Purpose	Assertion
<code>cgl_transform_check</code> <code>cgl_lf_quant_parallel</code>	$A \leftrightarrow \mu_B$ and primitive recovery sinusoidal $T_{\parallel}$ decay	pass final Fourier amplitude within 5% of analytic decay
<code>cgl_lf_quant_parallel</code> <code>_collisional</code>	sinusoidal $T_{\parallel}$ decay with $\nu_c > 0$	final Fourier amplitude within 7% of finite-collision analytic decay
<code>cgl_lf_quant_perp</code> <code>cgl_lf_quant_perp</code> <code>_collisional</code>	sinusoidal $T_{\perp}$ decay sinusoidal $T_{\perp}$ decay with $\nu_c > 0$	final Fourier amplitude within 5% of analytic decay final Fourier amplitude within 7% of finite-collision analytic decay
<code>cgl_lf_quant_grad_b</code>	isolated $\nabla_{\parallel} B $ term	RMS relative error below 10%, positive response correlation
<code>cgl_lf_flux_limiter</code> <code>cgl_lf_limiter</code> <code>_heat_flux_suppression</code>	equation 3.2 heat-flux cap limiter-collision suppression of heat fluxes	$ q \leq q_{\max}$, sign preserved, update within 20% active limiter scattering re- duces the unlimited LF flux by more than 20x
<code>cgl_lf_limiter_mirror</code>	mirror-biased admissibility stress	intermediate STS states fi- nite, positive, and backup bounded
<code>cgl_lf_limiter_firehose</code>	firehose-biased admissibility stress	intermediate STS states fi- nite, positive, and backup bounded
<code>cgl_lf_field_wave</code>	field-aligned LF wave reference	complex Fourier mode within 25% of offline reference
<code>cgl_lf_paper_oblique_wave</code>	Figure 17 background, LF en- abled	complex Fourier mode within 40% of offline reference
<code>cgl_pure_paper_oblique_wave</code>	Figure 17 background, no LF	complex Fourier mode within 40% of offline reference

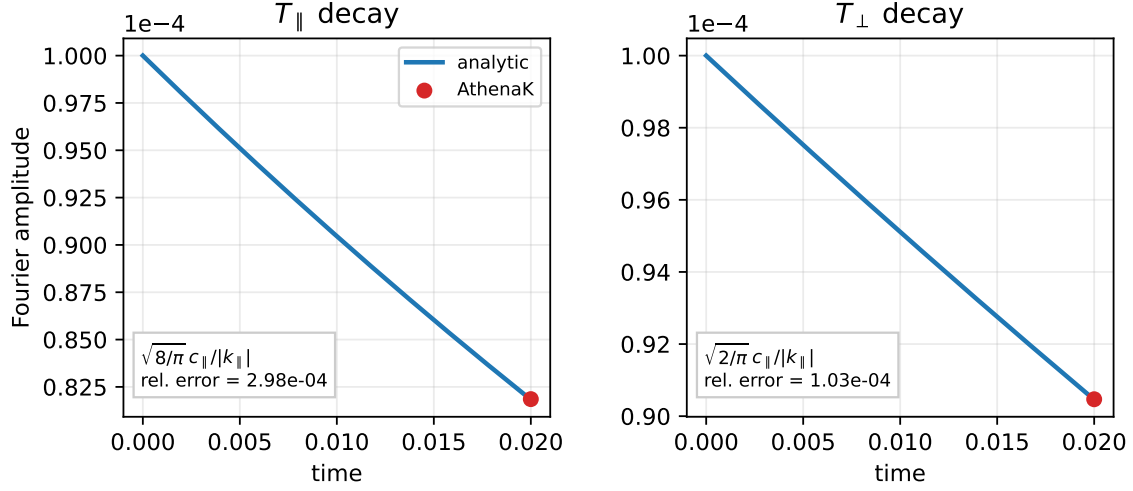


Figure 1: Constant-coefficient decay tests. The $T_{\parallel}$ case uses $\chi_{\parallel} = \sqrt{8/\pi} c_{\parallel}/|k_{\parallel}|$; the $T_{\perp}$ case uses $\chi_{\perp} = \sqrt{2/\pi} c_{\parallel}/|k_{\parallel}|$. The red point is the final AthenaK Fourier amplitude and the blue curve is the analytic constant-coefficient decay.

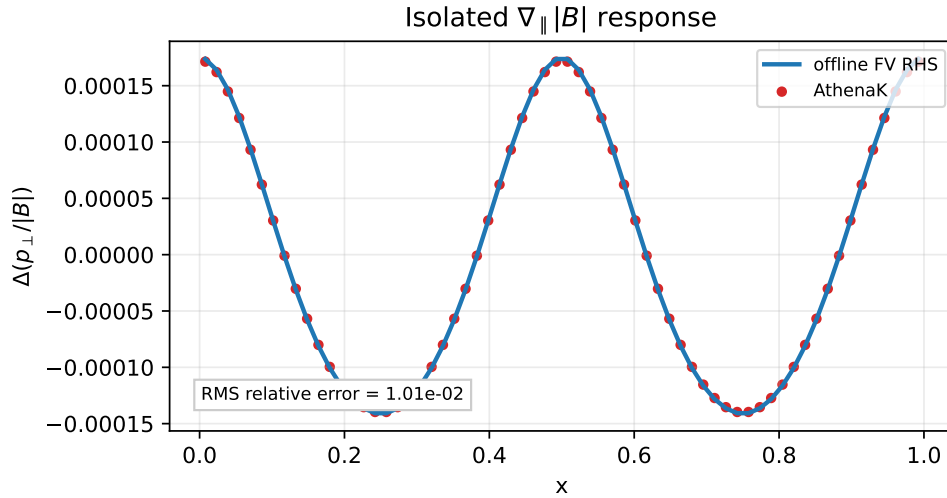


Figure 2: Isolated $\nabla_{\parallel} |B|$ term in the perpendicular heat flux. The initial thermal gradients are zero; the plotted response is the short-time change in $p_{\perp}/|B|$ compared with an offline finite-volume reconstruction of the initial LF RHS.

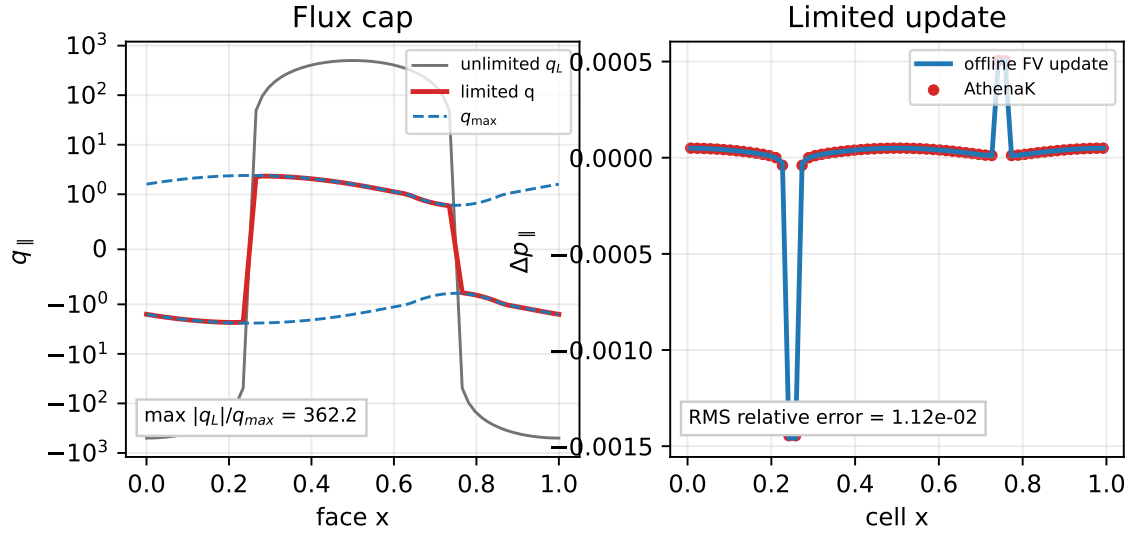


Figure 3: Heat-flux limiter test. The left panel shows the unlimited $q_{||,L}$, the capped $q_{||}$, and $\pm q_{||,max}$. The right panel checks that the pressure update produced by the limited flux matches the offline finite-volume update.

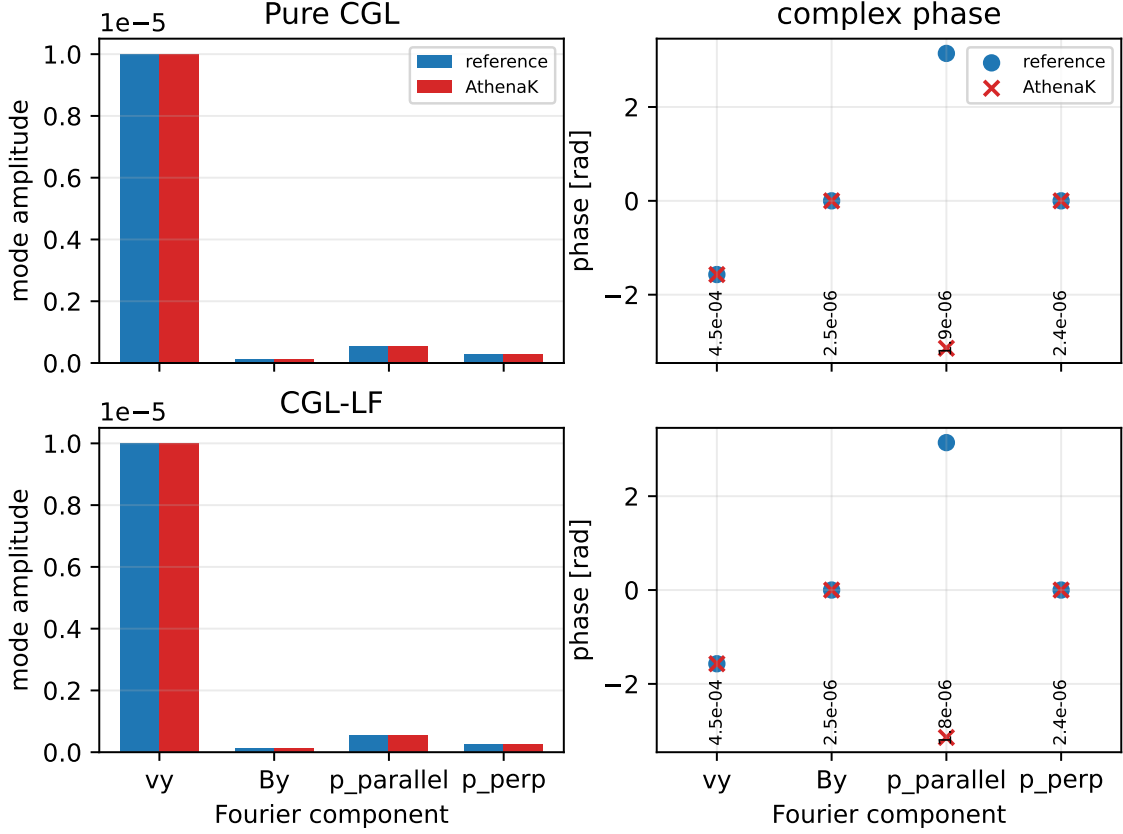


Figure 4: Figure 17-style oblique linear-wave check. The plotted quantities are the final complex Fourier amplitudes of u_y , B_y , $p_{\parallel}$, and $p_{\perp}$ for the pure-CGL and CGL-LF inputs, compared with the offline linearized reference used by the quantitative pgen. The vertical phase-panel labels are per-variable relative errors.

8 Figure 17-Style Linear Wave

The paper-style inputs use the Figure 17 background

$$\rho_0 = 1, \quad p_{\parallel 0} = p_{\perp 0} = 5, \quad u_0 = 0, \quad B_0 = (1, \sqrt{2}, 0.5), \quad k = 2\pi\hat{x}, \quad kL = |k|. \quad (9)$$

The test initializes a small transverse velocity perturbation and compares the final complex Fourier amplitudes of u_y , B_y , $p_{\parallel}$, and $p_{\perp}$ against an offline linearized CGL–LF reference integrated with the same background state and closure coefficients. Two inputs are provided: one with the LF heat flux enabled and one with pure CGL. This checks the same method class as the Appendix A.2/Figure 17 linear-wave convergence problem, while leaving exact eigenmode initialization and full convergence scans for a follow-up once Steve/Jono provide canonical eigenvectors or damping rates.

9 Build and Runtime Status

The CPU quantitative build completed successfully. The only build warnings observed were pre-existing variable-length-array warnings in `mesh.cpp`. No CGL-owned files were deleted.

References

- [1] J. Squire et al., *Pressure anisotropy and viscous heating in weakly collisional plasma turbulence*, arXiv:2303.00468, 2023.